

Service REST is Configured to Execute Asynchronously.

- Service Rest rule with Execution Mode as Execute Asynchronously (queue for execution) and Service Request Processor

The screenshot shows the 'Service REST: TestAsynchronous' configuration window. The 'Resource properties' section includes the 'Service endpoint URL' set to 'https://rkulcd0l.pegace.net/prweb/api/RSPackage/V1/CustomerDetailAsync'. The 'Processing options' section shows the 'Execution mode' set to 'Execute asynchronously (queue for execution)' and the 'Request processor' set to 'TestAsync'.

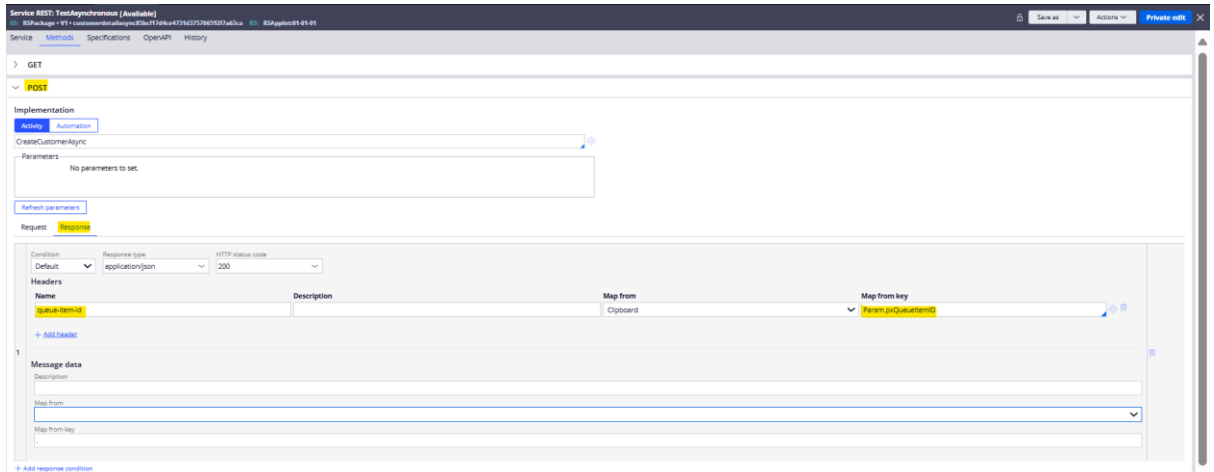
- Service Request Processor with queue class: **System-Queue-ExecutionRequest-Service-Default**

The screenshot shows the 'Edit Service Request Processor: TestAsync' configuration window. The 'Queuing Options' tab is active, showing a table with one entry: 'System-Queue-ExecutionRequest-Ser'. The 'Dequeueing Options' tab is also visible, showing 'Maximum number of execution attempts' set to 1, and checkboxes for 'Keep in queue after all execution attempts failed' and 'Keep in queue after successful execution'.

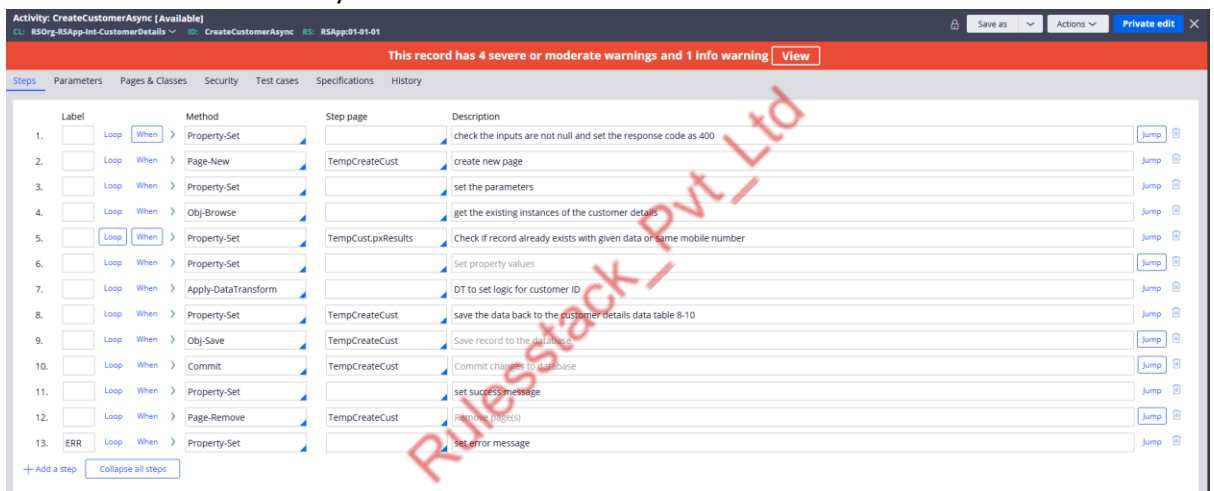
- Request Tab of Post Method and Service Activity

The screenshot shows the 'Service REST: TestAsynchronous' configuration window, specifically the 'Request' tab for the 'POST' method. The 'Implementation' section shows the 'Activity' set to 'CreateCustomerAsync'. The 'Parameters' section is empty. The 'Headers' section is empty. The 'Resource path parameters' section is empty. The 'Query string' section is empty. The 'Message data' section shows the 'Map to' set to 'JSON' and the 'Map to key' set to 'ReqPage'.

- Response tab of Post method, Queue Item Id mapped in Response Header



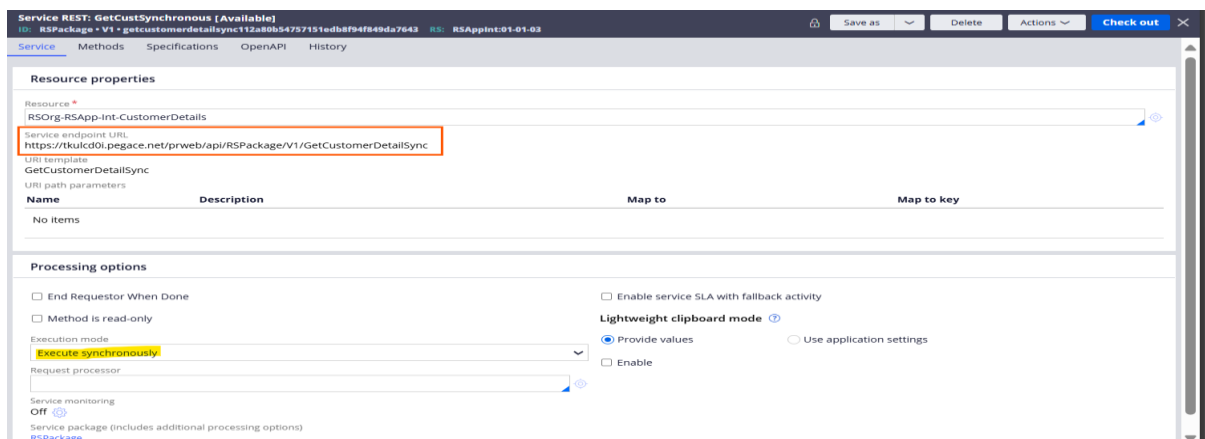
- Post Method Service Activity which will create Customer ID



Above rules are triggered when Service hit happens to create Customer ID using Post Method.

- When Service hit happens, Service will Queue the instance to Queue Class which is mentioned in Service Request Processor and send the Queue ID in Response header and run Asynchronously.

To get the Created Customer ID from data table another Service hit happens. We can see the Get Method in another Service Rest.



- Passing Queue ID in Request Header and using OOTB Activity which requires Queue ID to get the Generated Response.

Service REST: GetCustSynchronous [Available]
ID: RSPackage - V1 - getcustomerdetailsync112a80b54757151ed8f94849da7643 RS: R5Apptm:01-01-03

Service Methods Specifications OpenAPI History

GET

Implementation
Activity Automation

GetExecutionRequest

Parameters
ItemID Param.ItemID

Refresh parameters

Request Response

Headers
Name Description Map to Map to key
queue-item-id Description Clipboard Param.ItemID

+ Add header

Resource path parameters
Name Description Map to Map to key
No items

Query string

- Response

Service REST: GetCustSynchronous [Available]
ID: RSPackage - V1 - getcustomerdetailsync112a80b54757151ed8f94849da7643 RS: R5Apptm:01-01-03

Service Methods Specifications OpenAPI History

GET

Implementation
Activity Automation

GetExecutionRequest

Parameters
ItemID Param.ItemID

Refresh parameters

Request Response

Condition Response type HTTP status code
Default application/json 200

Headers
Name Description Map from Map to key

+ Add header

Message data
Description
Map from JSON
Map to key ResPage
Mapping method Use legacy method

+ Add response condition

- Get Service Activity **GetExecutionRequest** (OOTB) Retrieves Execution Requests from queues derived from the System-Queue-ExecutionRequest- class and provides the response with respective Customer ID

Activity: GetExecutionRequest [Available]
CL: @baseclass ID: GetExecutionRequest RS: Pega-RULES-08-01-01

This record has 2 severe or moderate warnings (including 2 unjustified) View

Steps Parameters Pages & Classes Security Test cases History

1. Loop When Java Description Java Jump

Method Parameters

Java Source:

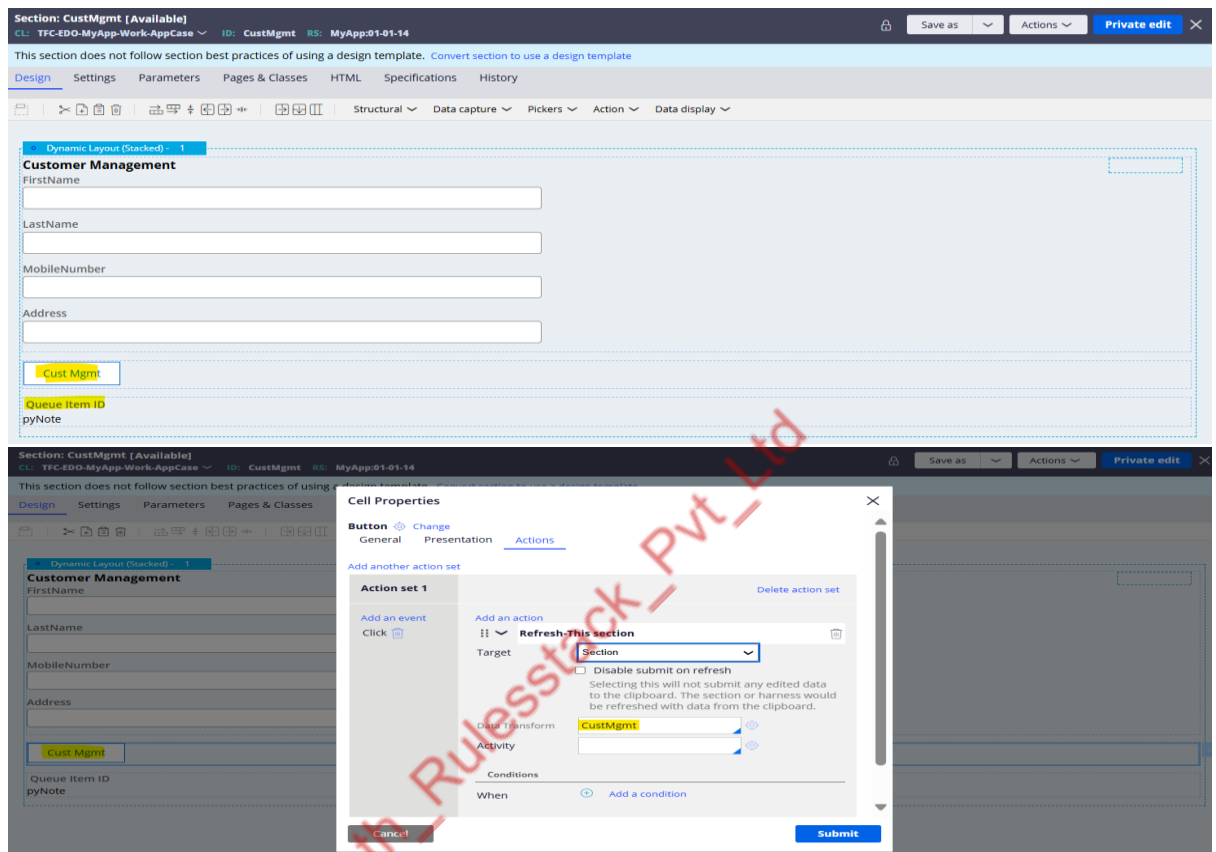
```

1 String ItemID = tools.getParamValue("ItemID");
2 com.pegarules.pub.util.HashMap options =
3 new com.pegarules.pub.util.HashMap();
4 options.put(PRIQueueManager.OPTION_UPDATE_STATS, "false");
5 try {
6 tools.getThread().getRequestProcessor().dequeueExecutionRequest(ItemID,
7 (com.pegarules.pub.util.HashMap)options);
8 }
9 catch (Exception ex){
10 String errorMessage = "Failed to dequeue Item: " + ItemID + ", " + ex.toString();
11 olog.error(errorMessage, ex);
12 tools.putParamValue("pxExecutionStatus", "");
13 tools.putParamValue("pxErrorMessage", errorMessage);
14 }
15 if (olog.isDebugEnabled()){
16 olog.debug(tools.getPrimaryPage().getXML());
17 olog.debug(tools.getParameterPage().toString());
18 }

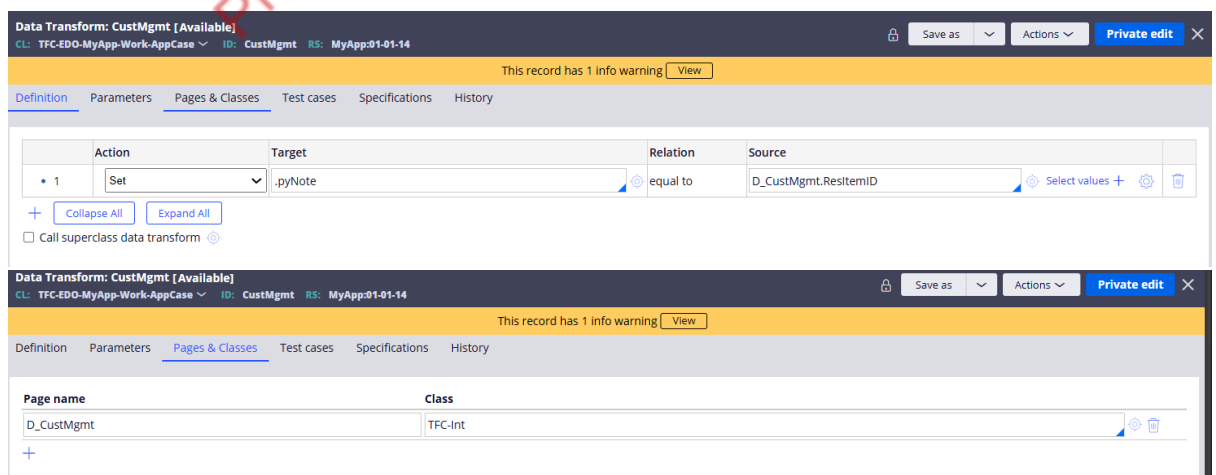
```

Consuming the Post Service which executes Asynchronously.

- On click of button in UI, Data Transform is used to Trigger the Post Service which will Send the Queue ID in Response Header.
- On Click of button, Service hit happens and Queue ID received is displayed in “Queue Item ID”.



Data Transform which runs on click of button



- Data Page with Post Method Source which runs Connect Rest to get the Queue ID

Data Page: CustMgmt [Available] Save as Actions Private edit

Service: TFC-Int-CustMgmt MyAppln01 01-04

This record has 1 severe or moderate warning and 1 info warning [View](#)

Definition Load Management Parameters Pages & Classes Test cases Usage Specifications History

Data page definition

Structure

Page ▼

Object type ▼

TFC-Int

Mode ▼

Read-Only

☐ Restricted feature set for high throughput [?](#)

Scope ▼

Thread

Data Sources

If there are multiple sources, the first IF that evaluates to true is used. If none evaluates to true, the OTHERWISE source is used.

1 **IF** CheckQueueID Parameters

Source ▼ Connector Type ▼ REST Name ▼ GetCust TFC-Int-GetCustomerParameters

Request Data Transform ▼ GetReqDT Parameters

Response Data Transform ▼ GetResDT Parameters

Endpoint URL ▼ Method ▼ Get

☐ Simulate data source

☐ Run on data page

2 **OTHERWISE**

Source ▼ Connector Type ▼ REST Name ▼ CreateCust TFC-Int-CreateCustomerParameters

Request Data Transform ▼ PostReqDT Parameters

Response Data Transform ▼ PostResDT Parameters

Endpoint URL ▼ Method ▼ Post

☐ Simulate data source

☐ Run on data page

[Add alternative source](#)

Connect REST: CreateCust [Available] Save as Actions Private edit

Service: TFC-Int-CreateCustomer MyAppln01 01-04

Service Methods Parameters Specifications History

Get

Post

Request Response

Headers

Name	Description	Map from	Map from key
		▼	▼

Additional query string parameters

Name	Description	Map from	Map from key
		▼	▼

Message data

Message type ▼ Default ▼ www-form-urlencoded

Description

Map from ▼ JSON

☐ Use fast processing

Map from key ▼ PostReqDT

☐ Call is read-only

Connect REST: CreateCust [Available] Save as Actions Private edit

Service: TFC-Int-CreateCustomer MyAppln01 01-04

Service Methods Parameters Specifications History

Get

Post

Request Response

Headers

Name	Description	Map to	Map to key
queue-item-id		Clipboard	RestItemID

Message data

Description

Map to ▼

Map to key

Data Transform: PostReqDT [Available] Save as Actions Private edit

Service: TFC-Int-CreateCustomer MyAppln01 01-04

This record has 1 info warning (including 1 unjustified) [View](#)

Definition Parameters Pages & Classes Test cases Specifications History

	Action	Target	Relation	Source
1	Update Page	PostReqPg	with values from	pyWorkPage.CustDetail
1.1	Set	FirstName	equal to	FirstName
1.2	Set	LastName	equal to	LastName
1.3	Set	MobileNumber	equal to	MobileNumber

[Collapse All](#) [Expand All](#)

Data Transform: PostReqDT [Available]

TFC-Int-CreateCustomer | PostReqDT | MyAppInt01-01-04

This record has 1 info warning (including 1 unjustified) [View](#)

Definition Parameters Pages & Classes Test cases Specifications History

Page name: pyWorkPage Class: TFC-EDO-MyApp-WorkAppCase

Data Transform: PostReqDT [Available]

TFC-Int-CreateCustomer | PostReqDT | MyAppInt01-01-04

This record has 1 info warning (including 1 unjustified) [View](#)

Definition Parameters Pages & Classes Test cases Specifications History

Action	Target	Relation	Source
1 When	pxDataPageHasErrors		
1.1 Comment	Call template Data transform for automatic error handling of data pages		
1.2 Apply Data Transform	pxErrorHandlerTemplate		
2 Update Page	Primary	with values from	DataSource
2.1 Set	RestHeader	equal to	RestHeader

[Collapse All](#) [Expand All](#)

- Loading of second screen
- Flow Action Pre Processing Data Transform, it calls Data Transform at second screen loading time.

Flow Action: CustMgmtDis [Available]

TFC-EDO-MyApp-WorkAppCase | CustMgmtDis | MyAppInt01-01-14

Layout Validation Action Help setup Security HTML Pages & Classes Specifications History

Pre-processing

Occurs before this action is loaded

Apply data transform

CustMgmtDis [Refresh parameters](#)

Parameters

No parameters to set.

Run activity

Run robotic automation Description [Refresh parameters](#)

Post-processing

- Data Transform which calls Data Page for second service hit

Data Transform: CustMgmtDis [Available]

TFC-EDO-MyApp-WorkAppCase | CustMgmtDis | MyAppInt01-01-14

This record has 1 info warning (including 1 unjustified) [View](#)

Definition Parameters Pages & Classes Test cases Specifications History

Action	Target	Relation	Source
1 Update Page	Primary	with values from	D_CustMgmt
1.1 Set	CustomerID	equal to	CustID
1.2 Set	pyMessage	equal to	Msg

[Collapse All](#) [Expand All](#)

- Data Page with Get Method Source which runs Connect Rest to get the Customer ID as Response.

Data Page: CustMgmt [Available]

D_CustMgmt | MyAppInt01-01-04

This record has 1 severe or moderate warning and 1 info warning [View](#)

Definition Load Management Parameters Pages & Classes Test cases Usage Specifications History

Data page definition

Structure: Page

Object type: TFC-Int

Mode: Read-Only

Scope: Thread

Data Sources

If there are multiple sources, the first IF that evaluates to true is used. If none evaluate to true, the OTHERWISE source is used.

IF [CheckQueueID](#) [Parameters](#)

Source	Type	Name	Request Data Transform	Response Data Transform	Endpoint URL	Method
1 Connector	REST	GetCust	GetReqDT	GetResDT		Get

☐ Run on data page

OTHERWISE

Source	Type	Name	Request Data Transform	Response Data Transform	Endpoint URL	Method
2 Connector	REST	CreateCust	PostReqDT	PostResDT		Post

☐ Run on data page

[Add alternative source](#)

- Data Page Sources get triggered based on Condition Satisfied

When: CheckSummed [Available]
CL: TFC-Int-GetCustomer ID: GetCust RS: MyAppInt:01-01-04

Conditions Advanced Parameters Pages & Classes Test cases Specifications History

This record has 1 info warning (including 1 unjustified) View

H 1 pyWorkPage.pyNote Select values

Add condition

Logic string

(1)

Display property labels

Advanced options

Allowed functions

No items

Add function

Skip backward chaining

Connect REST: GetCust [Available]
CL: TFC-Int-GetCustomer ID: GetCust RS: MyAppInt:01-01-04

Save as Actions Private edit

Service Methods Parameters Specifications History

Get

Request Response

Headers

Name	Description	Map from	Map from key
.queue-item-id		Clipboard	.ReqItemID

Additional query string parameters

Name	Description	Map from	Map from key

Connect REST: GetCust [Available]
CL: TFC-Int-GetCustomer ID: GetCust RS: MyAppInt:01-01-04

Save as Actions Private edit

Service Methods Parameters Specifications History

Get

Request Response

Headers

Name	Description	Map to	Map to key

Message Data

Description

Map to

JSON

Use fast processing

Map to key

.GetResPg

Data Transform: GetReqDT [Available]
CL: TFC-Int-GetCustomer ID: GetReqDT RS: MyAppInt:01-01-04

Save as Actions Private edit

Definition Parameters Pages & Classes Test cases Specifications History

This record has 1 info warning (including 1 unjustified) View

Action	Target	Relation	Source
1 Set	.ReqItemID	equal to	pyWorkPage.pyNote

+ Collapse All Expand All

Call superclass data transform

Data Transform: GetResDT [Available]
CL: TFC-Int-GetCustomer ID: GetResDT RS: MyAppInt:01-01-04

Save as Actions Private edit

Definition Parameters Pages & Classes Test cases Specifications History

This record has 1 info warning (including 1 unjustified) View

Action	Target	Relation	Source
1 Update Page	Primary	with values from	DataSource.GetResPg
1.1 Set	.CustID	equal to	.CustomerID
1.2 Set	.Msg	equal to	.pyMessage
2 When	pxDataPageHasErrors		
2.1 Comment	Call template Data transform for automatic error handling of data pages		
2.2 Apply Data Transform	pxErrorHandlingTemplate		

+ Collapse All Expand All

Call superclass data transform

Functionality Testing for the above Use Case

- User Entered the details

The screenshot shows the 'AppCase' form for case A-24006. The form is in the 'Create' stage, with 'Stage 2' and 'Stage 3' also visible. The 'CustMgmt' button is highlighted. The form fields are as follows:

Field	Value
FirstName	shiva
LastName	prasad
MobileNumber	8520034561
Address	hyd

Buttons: Cust Mgmt, Cancel, Save, Submit.

Metadata:

Field	Value
EmailFrom	--
EmailSubject	--
Body	--
CustomerID	--
Case ID	A-24006

- On click of Cust Mgmt button, first service hit happens and Queue id is displayed in UI

The screenshot shows the 'AppCase' form for case A-24006. The 'CustMgmt' button has been clicked, and the 'Queue Item ID' is displayed in a red box. The form fields are as follows:

Field	Value
FirstName	Shiva
LastName	Prasad
MobileNumber	8520034561
Address	HYD

Buttons: Cust Mgmt, Cancel, Save, Submit.

Queue Item ID: SYSTEM-QUEUE-EXECUTIONREQUEST-SERVICE-DEFAULT 1745561536808000PEGA-PEGA-85C956FD6D-VFMV9

Metadata:

Field	Value
EmailFrom	--
EmailSubject	--
Body	--
CustomerID	--
Case ID	A-24006

- On click of submit button, second service hit happens and displays the customer id and message

The screenshot shows the 'AppCase' form for case A-24006. The 'Submit' button has been clicked, and the 'CustomerID' and 'pyMessage' are displayed in red boxes. The form fields are as follows:

Field	Value
CustomerID	250420251005P
pyMessage	Successfully Created New Customer ID

Buttons: Cancel, Save, Submit.

Metadata:

Field	Value
EmailFrom	--
EmailSubject	--
Body	--
CustomerID	250420251005P
Case ID	A-24006